# Cognitive Modeling

## Complete Lecture Notes

Week 16: Final Project Presentations and Good Modeling Practice

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# Chapter 1

# Applications in Cognitive Science

## 1.1 Week 16: Final Project Presentations and Good Modeling Practice

> Course wrap-up · Lecture slot: synthesis + first presentations · Lab slot: presentations and peer feedback

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### 1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

Table 1.1: Key terms

| English | فارسی | توضیح کوتاه |
|---|---|---|
| peer feedback | بازخورد همتا | نقد سازنده‌ی کار هم‌کلاسی‌ها |
| fit / fitting | برازش | تنظیم پارامترها تا مدل با داده جور شود |
| parameter recovery | بازیابی پارامتر | آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند |
| reproducibility | بازتولیدپذیری | امکان به دست آوردن دوباره‌ی همان نتایج |
| simulation | شبیه‌سازی | تولید داده‌ی مصنوعی با اجرای مدل |
| estimate / estimator | برآوردگر / برآورد | روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن |
| parameter | پارامتر | کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند |

| English | فارسی | توضیح کوتاه |
|---|---|---|
| likelihood | درست‌نمایی | احتمال داده به‌عنوان تابعی از پارامترها |
| Bayes factor (BF) | عامل بیز (Bayes factor) | نسبت درست‌نمایی‌های حاشیه‌ای دو مدل |
| bounds / constraints | قیدها / کران‌ها | محدودیت‌های مجاز برای مقدار پارامترها |
| posterior | توزیع پسین (posterior) | باور درباره‌ی پارامتر بعد از دیدن داده |
| hierarchical model | مدل سلسله‌مراتبی (چندسطحی) | پارامترهای فردی از یک توزیع گروهی می‌آیند |
| adequacy check | بررسی کفایت مدل | آیا مدل ویژگی‌های اصلی داده را بازتولید می‌کند |
| posterior predictive check (PPC) | بررسی پیش‌بینی پسین | مقایسه‌ی داده‌ی شبیه‌سازی‌شده از پسین با داده‌ی واقعی |
| cross-validation (CV) | اعتبارسنجی متقابل (cross-validation) | برازش روی بخشی از داده و آزمون روی بخش کنارگذاشته |
| AIC / BIC | بیزی / معیار اطلاعاتی آکائیکه | برازش به‌علاوه‌ی جریمه برای تعداد پارامتر |
| research design | طرح آزمایش | شرایط، تعداد کوشش‌ها و نمونه‌ها |
| model recovery | بازیابی مدل | آیا مقایسه‌ی مدل، مدل مولد را درست انتخاب می‌کند |
| drift-diffusion model (DDM) | مدل drift-diffusion (رانش–انتشار) | انباشت پیوسته‌ی شواهد بین دو مرز |
| preregistration | پیش‌ثبت (preregistration) | ثبت برنامه‌ی تحلیل پیش از دیدن داده |
| prior | توزیع پیشین (prior) | باور درباره‌ی پارامتر پیش از دیدن داده |
| random seed | seed تصادفی | عدد شروع مولد اعداد تصادفی برای تکرارپذیری |
| large language model (LLM) | مدل بزرگ زبانی | شبکه‌ی عصبی‌ای که کلمه‌ی بعدی متن را پیش‌بینی می‌کند |
| accuracy | دقت | نسبت پاسخ‌های درست |

### 1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Communicate a complete modeling study (question, model, fitting, evaluation, recovery, interpretation) clearly and within a time limit.
2. Critique other modeling studies constructively, using the course's evaluation criteria.
3. Apply a checklist of good modeling practice (reproducibility, transparency, robustness) to their own project report.
4. Integrate the methods of the course into one coherent workflow and explain when to use which.

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### 1.1.3 Motivation & big picture

A model is only as useful as others' ability to understand, check and build on it. Over 15 weeks, every lab has followed the same arc: formalize → simulate → fit → evaluate → recover → interpret. This week you show that arc in your own project, and you practice the other half of scientific modeling: reading someone else's model critically but fairly.

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### 1.1.4 Lecture notes

In plain words The last week has no new method. Students present their projects, and the course is summarized as one workflow: formalize a theory as a model, simulate it to see what it predicts, fit it to data, evaluate it (does it fit, and does it beat alternatives?), check that the method can recover the truth at this design, and interpret the result in psychological terms, including what cannot be concluded.

Suggested timing (lecture slot, 60 min): course synthesis 15 · good-practice checklist 15 · first presentations 30.

#### 1. The course in one table

Table 1.2: 1. The course in one table

| Question you face | Tools | Week |
|---|---|---|
| How do I turn a theory into a model? | Five components; simulation; analytic checks | 1–2 |
| How do I estimate parameters? | Least squares; maximum likelihood; optimization (grid, Nelder–Mead, L-BFGS-B), multi-start, bounds | 3–4 |
| How certain are the estimates? | Hessian SEs, profile likelihood, bootstrap; posterior intervals | 5, 9 |
| How do I handle many participants? | Individual fits; aggregation artifacts; hierarchical models | 5, 13 |
| Is the model adequate? | Residuals, $G^2$, posterior predictive checks | 2–4, 10 |
| Which model is better? | CV, LRT, AIC/BIC, Bayes factors | 6–7, 10 |

| Question you face | Tools | Week |
|---|---|---|
| Can my design answer the question? | Parameter recovery, model recovery | 7 |
| How do I get a posterior for complex models? | Grid → Metropolis–Hastings → Gibbs → NUTS (); $\hat{R}$, ESS, divergences | 9–13 |
| What do models look like in different fields? | SDT, forgetting, practice, GCM, DDM/EZ, RL, ideal observers | 2, 5, 14–15 |

For an engineer Looking back, most of the course maps onto tools you already know, and the table below can serve as your own summary when answering students' questions:

Table 1.3: For an engineer

| Course topic | Engineering counterpart |
|---|---|
| Random walk / DDM (Weeks 1, 14) | Sequential detection (SPRT), first-passage times |
| SDT, ROC (Week 2) | Binary detection, $P_D$ / $P_{FA}$, ROC |
| Least squares, optimizers (Week 3) | LS estimation, gradient methods, Gauss–Newton |
| Maximum likelihood, $G^2$, LRT (Weeks 4, 7) | ML estimation, goodness-of-fit tests, GLRT |
| Hessian SEs, bootstrap (Week 5) | Fisher information / CRLB, Monte Carlo performance evaluation |
| CV, AIC, BIC (Weeks 6–7) | Model-order selection (AIC, MDL), validation data |
| Parameter/model recovery (Week 7) | Testing estimators on synthetic data with known ground truth |
| Bayesian inference, BF (Weeks 9–10) | MAP/MMSE estimation, Bayesian detection |
| MCMC, $\hat{R}$, ESS (Weeks 11–12) | Monte Carlo simulation, Markov processes, correlated-sample statistics |
| Hierarchical models (Week 13) | Empirical Bayes, James–Stein shrinkage |
| Delta rule, cue combination (Week 15) | LMS adaptation, MRC / Kalman sensor fusion |

What is genuinely different in cognitive modeling: the "system" is a human being, the data are small and noisy, and the parameters themselves are the scientific findings, so interpretation, uncertainty and recovery matter more than raw performance.

### 2. A checklist of good modeling practice

Before fitting - [ ] The research question is stated in terms of model parameters or model comparisons. - [ ] Candidate models (including at least one simpler alternative) are specified *before* looking at the

data, ideally preregistered. - [ ] Priors (if Bayesian) are justified and checked with prior predictive simulation. - [ ] Parameter and model recovery have been checked at the actual design.

Fitting and evaluation - [ ] The likelihood matches the data type (binomial for counts, sequential for learning, ...). - [ ] Optimization uses multi-start and sensible bounds, and fits are checked visually; for MCMC, $\hat{R}$, ESS and divergences are reported. - [ ] Absolute fit (residuals, $G^2$ or PPC) *and* relative fit (CV/IC/BF) are reported. - [ ] Uncertainty is reported for every interpreted parameter.

- [ ] Limitations are stated: what the design cannot distinguish, which parameters are not recoverable, which assumptions were not tested.

### 3. Presentation format

- Length: 8 minutes (individual) or 10 minutes (pairs), plus 4 minutes of questions. Strict time-keeping.
- Suggested structure (one slide each, roughly):

  1. The question and why it matters.
  2. The model(s): equations and the psychological meaning of every parameter.
  3. Data and design.
  4. Fitting method and a fit figure (data + predictions).
  5. Evaluation: model comparison *and* an adequacy check.
  6. Recovery: can the design answer the question?
  7. Interpretation and limitations.

- Pairs: both members present, and the Bayesian/MCMC or hierarchical component must be shown.
- Schedule: posted one week in advance; the order is drawn at random. Sessions run across the lecture and lab slots (about 15–18 presentations in 3 hours, depending on enrolment).

### 4. Giving and receiving feedback

Each student completes a peer-feedback form for three assigned presentations (below). Feedback is anonymous to the presenter, visible to the instructor, and counts toward participation (lab check-off). Good feedback is specific ("the recovery plot shows $\beta$ is not identifiable, so the group difference in $\beta$ should not be interpreted") rather than general ("nice talk"), and constructive (it suggests a concrete improvement).

Peer-feedback form

Table 1.4: 4. Giving and receiving feedback

| Item | Rating (1–4) | Comment |
|---|---|---|
| The question was clear and the model matched it | | |

| Item | Rating (1–4) | Comment |
|---|---|---|
| The model's parameters were explained in psychological terms | | |
| The fit was shown and evaluated (adequacy + comparison) | | |
| Recovery or identifiability was addressed | | |
| Conclusions followed from the results, and limitations were stated | | |
| One thing that was particularly well done | | |
| One concrete suggestion for the written report | | |

For the instructor

- Grade with the rubric, comment with the checklist. During presentations, keep the good-practice checklist next to you and note one strength and one concrete improvement per team; this makes the written feedback fast and fair.
- Questions to ask presenters (they map onto the course): Which parameter answers your question? How do you know it is recoverable? Does the model fit in absolute terms, not only better than the alternative? What can your design *not* tell us?
- Use the worked critique (fictional Q-learning study) as a model for how students should give feedback: specific, tied to a week of the course, and constructive.
- Close the course with the one-table summary. Ask students which row they found hardest, and point them to the corresponding week for exam revision.

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### 1.1.5 Worked example: critiquing a modeling study

Summary of a (fictional) study. *"We fitted a Q-learning model to 30 patients and 30 controls who performed 60 trials of a two-armed bandit task with stable reward probabilities. Each participant was fitted by MLE. Patients had significantly lower learning rates ($t(58) = 2.4$, $p = .02$). We conclude that the disorder impairs reward learning."*

Critique using the checklist.

1. Recovery at this design (Week 7, Week 15 lab). With 60 trials and *stable* reward probabilities, the learning rate is weakly identifiable, and $\alpha$ and $\beta$ trade off. Without a recovery study, a

difference in $\hat{\alpha}$ could reflect a difference in choice noise ($\beta$) or just estimation noise. *Suggestion:* report parameter recovery at this design, and include reversals.

2. Model comparison (Week 15, Part D). Did Q-learning beat simpler strategies (WSLS, perseveration) in both groups? If patients are better described by a different strategy, comparing their $\alpha$ with controls' is not meaningful. *Suggestion:* report AIC/BIC or CV per group, with model recovery.
3. Second-stage t-test on point estimates (Weeks 5, 13). It ignores the uncertainty of individual estimates, which is large with 60 trials. *Suggestion:* a hierarchical model with a group effect on $\alpha$, reporting its posterior.
4. Adequacy (Weeks 4, 10). No posterior predictive check or learning-curve plot is shown. Does the model reproduce each group's learning curve?
5. Interpretation. "Impairs reward learning" is a strong process claim. A lower $\hat{\alpha}$ is also compatible with more exploration, lower motivation (lower $\beta$), or different priors on initial values. *Suggestion:* temper the conclusion, or add a design that separates these accounts.

Overall: a reasonable first study whose conclusion is not yet supported. Three additions (recovery, model comparison, hierarchical estimation) would make it convincing. Notice that each point maps directly onto one week of this course.

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### 1.1.6 Lab (2 hours): presentations and peer feedback

Goals: present your project; give written feedback on three presentations; note the feedback you receive for your final report.

Logistics: - Upload slides (PDF) to the course page before the session starts. - Fill in the peer-feedback forms during the session (paper or online form) and submit them by the end of the day.

After the presentations: the instructor gives brief oral feedback to each team. Written feedback (instructor + peers) is returned within 3 days so it can be used in the final report.

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### 1.1.7 Connection to AI & LLMs

Good practice when AI is part of your modeling work. AI tools now enter cognitive modeling in three roles, and each needs the course's habits of checking: 1. 2. As a "simulated participant". Some studies give LLMs psychology experiments and compare the answers with human data, or even propose LLM-based models of human behavior (e.g., Binz & Schulz, 2023; Dillion et al., 2023 [verify]). Evaluate such claims like any model: generalization to new tasks, comparison with simpler cognitive models, mimicry, and interpretability (Week 15). 3. As an object of study. If you test an LLM, record the model name and version, the date, the exact prompts, the temperature and the number of samples. Without these, the study cannot be reproduced.

It looks plausible, and it even passes the case $c = 0$, but a recovery test shows that it gets the sign of $c$ wrong.

A liberal observer would be reported as conservative, which would reverse the conclusion of a bias study. The test with $c = 0$ alone would not have caught it.

- ⋆⋆ Write a similar recovery test for the EZ-diffusion function of Week 14, using the simulator from the same lab.

*Optional reading:* Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. *PNAS, 120*(6), e2218523120. Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences, 27*, 597–600 [verify].

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### 1.1.8 Common mistakes & misconceptions (in project presentations and reports)

1. Too much on the method, too little on the meaning. Every parameter and every result needs a psychological interpretation.
2. Showing a fit statistic without a fit figure. Always plot data against predictions.
3. Model comparison without an adequacy check, or vice versa.
4. Interpreting parameters that were not shown to be recoverable.
5. Overclaiming: "the model is correct", "the mind is Bayesian". State what the evidence supports and what it does not.
6. Running over time. Practice the talk; one idea per slide.

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### 1.1.9 Check your understanding (integrative review questions for the final exam)

1. A colleague has accuracy and RT data from a speed/accuracy manipulation and wants to know whether the manipulation affects caution or evidence quality. Outline a full modeling plan, naming the model, the fitting method, the evaluation, and the recovery check. Answer Model: drift-diffusion (EZ-diffusion for a first pass, or the full DDM by MLE / hierarchical Bayes). Fit per participant and condition. Compare models in which the manipulation affects only $a$, only $v$, or both (AIC/BIC or a hierarchical model with condition effects; PPC on RT quantiles). Recovery: simulate data at the actual design with each model, and check that parameters and the "which parameter changed" conclusion are recovered. Interpret the condition effects on $a$ and $v$ with uncertainty.

2. When would you choose a Bayes factor over cross-validation for comparing two models, and vice versa? Answer Bayes factor: when the question is which model is more probable given the data (evidence, including support for a null), when meaningful priors are available, and with a prior sensitivity analysis. CV: when the goal is predictive accuracy for new data of a specific kind, priors are hard to justify, or the BF is too sensitive to the priors.

3. Explain how a parameter can be estimated very precisely (narrow CI or posterior) and yet be meaningless. Answer Precision only reflects the curvature of the likelihood under the assumed

model. If the model is misspecified (fails PPC/$G^2$), if the parameter trades off with another so that its meaning depends on an arbitrary constraint, or if the optimizer or sampler failed (local minimum, bound, non-convergence), the precise estimate does not measure the intended construct.

4. Why do hierarchical models often give better individual estimates than separate individual fits? Answer They share information across participants through the group distribution, shrinking noisy individual estimates toward the group mean in proportion to their uncertainty. This reduces estimation variance more than it adds bias, so estimates are on average closer to the truth.

5. Name three different ways in which this course used simulation, and why each matters. Answer (i) Exploring a model's predictions before fitting (what can and can't it produce). (ii) Parameter and model recovery (can the design and method answer the question). (iii) Bootstrap and predictive checks (uncertainty and adequacy). Further: Monte Carlo and MCMC for computing posteriors.

6. A model has the best AIC among four candidates but a posterior predictive check shows a systematic misfit at long delays. What do you report and conclude? Answer Report both. The model is the best of the candidates for prediction, but it is not adequate in absolute terms: none of the candidates captures the long-delay behavior. Conclusions should be restricted accordingly, and a model revision targeting that feature is the next step.

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### 1.1.10 Assignment

#### Final project: written report

- Due: one week after this session, and in any case before the final exam (exact date on the course page).
- Use the feedback from today: the rubric rewards visible responses to presentation feedback.

#### Final exam

Written, cumulative (emphasis on Chapters 4–6), formula sheet provided, no laptops; worth 3 of 20 points.

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### 1.1.11 Further reading

- Farrell & Lewandowsky (2018): the chapter on using models in psychology (the role of models in theory development and common pitfalls) [verify chapter number].
- Lee & Wagenmakers (2014): browse the case studies in Part IV for ideas on how to extend your project.
- Martin, Kumar & Lao (2021), chapter 9. End-to-end Bayesian workflows, a good model for structuring a report.

- *Optional, outside the course list:* Wilson, R. C., & Collins, A. G. E. (2019). Ten simple rules for the computational modeling of behavioral data. *eLife, 8*, e49547.

Thank you for a great course, and keep modeling!